

Allusions to Grandeur

Engineering handover | 11 September 2026 | P:\GameDev\AllusionsTooGrandeur | branch
codex/game-planning

Read this, then **AGENT_BRIEF.md** and **ROADMAP.md** in the repo root. The brief is the full cold handover and ROADMAP.md is the authoritative work queue. This PDF is a snapshot of where things stand and what was learned getting here — it goes stale, the repo does not.

1. What the game is

A dense, violent, surreal living-world sandbox in Godot 4.7. Not open-world-large — small, interconnected, heavy with consequence. Closer to Kenshi or Fallout than GTA. The world existed before the player and remembers what the player does to it. First region is **The Ashbloom Expanse**: post-nuclear Australian wasteland, runaway fungal ecology, decayed cybernetics, raider roads, buried anatomical industry.

Tone: grimy, crude, darkly funny. Postal 2's deadpan and early South Park's absurdism, with Cruelty Squad's body modification and Wrought Flesh's organ-forward biopunk. A disembowelling and a petty argument about parking in the same thirty seconds. Satire aims at institutions and power, never at real groups.

Greg owns art direction, world, narrative and every final decision. You are an engineering partner, not a creative replacement. Offer an opinion once, clearly, then build what he decides.

2. Non-negotiables

- **Originality over imitation.** Every named reference — Nemesis System, Half Sword, Dark Souls, BeamNG, Sniper Elite — is an *experiential* reference only. Never extract or reproduce assets or protected implementations. **Warner Bros holds US Patent 10,926,179 on the Nemesis System**, in force into the mid-2030s. A deliberate reproduction is the most legally exposed thing this project could build, and the game carries Greg's real name and art identity.
- **Systems create stories.** Never hand-author what the simulation could produce.
- **Bodies remember.** Injuries, prosthetics and scars persist and change behaviour.
- **Reciprocity.** Anything the player can do to another person can be done to the player. This is now a load-bearing pillar, not a nice-to-have (see §7).
- **One system at a time, proven in gameplay.** Do not attempt the whole game.
- **Never claim something works until you have verified it.** Never claim a visual result you have not looked at. This is the rule that catches the most mistakes — see §6.

3. Architecture you must respect

The event spine

systems/world_history.gd, autoloader as **WorldHistory**. Systems do not wire into each other; they record events and read subjects. A collision must be able to create body damage, a grudge, a repair demand, a rumour on the in-game internet and a Compendium entry **without any of those features knowing about each other**. If you are adding a direct call from one gameplay system into another, stop and route it through an event.

register_subject() creates or *migrates* — new fields merge into saved subjects without erasing earned grudges, wounds or bonds. Preserve that property in every change.

The vehicle contract

systems/arcade_vehicle.gd is a force-based rigid body. Engine and tire forces never replace the body's transform or velocity. Writing **linear_velocity** directly discards the solver's collision response and makes every impact weightless. A prior agent broke this for the AI cars and it silently ruined the derby for weeks.

The body

systems/baseline_human.gd — one rig for every person in the world: the player, derby drivers, hunted NPCs. It owns the zone vocabulary, the hit geometry, the skeleton, the organs and the anatomy state together. **systems/body_mesh.gd** generates its geometry. **Do not build a second kind of body.** That fragmentation is exactly what was just removed.

The look

systems/world_look.gd centralises Environment and material construction behind named presets (bone_yard, ashbloom, ossuary). Do not hand-roll an Environment in a scene — every scene that did rendered nearly black, and all of them have now been migrated.

4. What exists — do not rebuild

- **Opening:** vat_chamber.tscn — waking submerged, decanting, aisle walk, racked into a car. Works.
- **Derby:** rift_derby.tscn — 12 physical AI wreckers on the player's chassis, staged part shedding, crowd, win/loss, hand-off to Ashbloom.
- **Impact feel:** separation, contact lockout, blame attribution, first-hit destruction, camera shake.
- **Bodies:** proportioned mesh, skeleton (skull, spine, ribcage, long bones), seven organs with individual consequences, compound fractures, physical severed limbs, a downed state with execute / behead / spare.
- **Hunt Grounds:** bone_yard_hunt.tscn — first/third person, melee resolved by where you aim, encounter actors, loot, district generation, Reality Misfires.
- **Kill cam:** systems/kill_cam.gd — X-ray plate, shockwave, ribs fracturing in sequence, organs rupturing on delays. **Largely built and under-used** — it is the Sniper Elite register Greg wants and it is waiting for a trigger.
- **Dossier:** systems/character_archive.gd — Living Kinship Web, X-ray scan, Tree alignment.
- **Handheld:** systems/handheld_device.gd — Index / Map / Tree / Wire / Carry. This is the “phone” the mind-stamp asset list should live on.
- **Audio:** positional engine layers, impact pool, crowd, quarry reverb. Waveforms are generated and provisional.

5. What was fixed on 11 Sep, and why it matters

Each of these was found by measuring rather than reading. They are listed because the same class of fault is probably still present elsewhere.

Symptom	Actual cause
Rams felt weightless	Tire grip was an uncapped lateral spring — a car T-boned at 8 m/s generated 40 m/s² of counter-force and cancelled
Wreckers never engaged	The chassis scales steering authority by speed, so a stationary car cannot turn; the AI set throttle to zero when s
Wounds landed nowhere	Zone names disagreed across call sites (“legs”, “left arm”) and AnatomyComponent silently resolved anything un
Menu looked broken	Hand-rolled Environment, flat 0.5 ambient, one spotlight lighting a whole town. Also the colour setting wrote amb
Boom on boot	Engine layers started at driving volume; update_engine only runs once the round is active, so the whole countdo

The lesson to carry forward: in every one of these the written diagnosis was wrong and the measurement was right. Instrument before fixing, and correct the roadmap when the stated cause turns out to be false — a wrong cause recorded as fact costs the next agent more than the bug did.

6. How to verify — mandatory

Godot lives at **P:/GameDev/Tools/Godot-4.7.2/Godot_v4.7.2-stable_win64_console.exe**. Set **TEMP** and **TMP** to P:/GameDev/Temp and **ATG_TEST_MODE=1** so you never touch real saves.

Run the suites (all four must stay green):

```
--headless --path game res://tests/baseline_human_test.tscn --quit-after 5000
--headless --path game res://tests/opening_test.tscn --quit-after 5000
--headless --path game res://tests/impact_test.tscn --quit-after 5000
--headless --path game res://tests/derby_balance_test.tscn --quit-after 5000
```

Check a scene for runtime errors:

```
--headless --path game res://rift_derby.tscn --quit-after 400
```

Look at it — headless renders nothing:

```
--path game res://tests/capture_scene.tscn -- --scene=res://tests/body_showcase.tscn \
--out=P:/GameDev/Temp/look.png --frames=90
```

Then open the PNG and actually look at it. This caught detached arms, bone hidden inside opaque flesh, a near-black menu and a HUD collision — none of which any test failed on. **tests/body_showcase.tscn** exists purely for this: three bodies side by side, intact, wrecked and opened under the X-ray.

Note: single-script **--check-only** gives false positives for autoloads and for new **class_name** globals until the project is reimported with **--import**. FMOD, LimboAI and Terrain3D fail to copy their DLLs while the editor holds them open, so headless runs load without them. **FMOD currently fails to load and must not gate audio work.**

7. What to build next

ROADMAP.md is authoritative and more detailed. The headline is **Tier 1d — defeat, execution and subjugation**, the largest design in the project and captured whole on 11 Sep.

The pillar is reciprocity. Greg's framing: most games don't let the things you do to others happen to you. The player can be gored, dismembered, fitted with bionics they did not choose, shackled, conscripted and mind-stamped.

The downed state is built and is the keystone — defeated but not dead, symmetric for player and NPC, with execute / behead / spare already on the rig. Everything below is a choice made inside that window. Build in this order:

- Resolution menu surfaced in gameplay in the downed window (the verbs exist; nothing calls them yet).
- Kill cam wired to the real zones and organs on execute — mostly an integration job.
- Player defeat routed to shackled rather than dead, plus deliberate death: forfeit carried loot and be re-decanted out of tar and nuclear oil. `vat_chamber.tscn` is already the birth scene.
- Mind-stamp and the asset list on the handheld — taskable, reachable, remotely executable.
- Targeted part selection in the VATS register, feeding the zone and organ geometry that now exists.
- Proximity voice: stage 1 (positional NPC speech) delivers the feel with no microphone work and is buildable on its own.

Known gaps, stated plainly

- AI wreckers cannot damage each other — only the player's chassis has its impact signal connected, so a third of the pit is theatre.
- The player carries both a coarse **health** int and a full rig. Unifying them means rebalancing the hunt loop, so it was recorded rather than done quietly.
- Derby cars are still smooth box silhouettes and read as karts. Blender work on `art/scrap_skiff_v1/`.
- Seven plugins installed, none referenced by any script. LimboAI is the right tool for the Hunt System rework and is unused.

8. How to work with Greg

- He fires several ideas in bursts, often mid-task. Treat each as captured intent, not a demand for immediate implementation. Build the one that unblocks the others and write the rest into ROADMAP.md so nothing is lost.
- Tell him plainly when something cannot or should not be built as described, give the real reason in a sentence or two, and offer the version you *would* build. He responds well to specifics and badly to hedging.
- Show evidence when you diagnose. “The nearest wrecker sat at exactly 12.6 m for thirty seconds” is useful. “The AI could be improved” is not.
- Send him screenshots. He judges by eye and he is usually right.
- Never add a key, toggle or reset whose purpose is to work around a broken system. He called this out directly: it “feels super ingenuine, like not a game”. Fix the system.
- Commit after every increment that runs. Git is the handoff protocol — this repo once sat at zero commits while three divergent copies of the same scripts accumulated.

Start by reading ROADMAP.md. Take the top item. Verify it with a capture you actually look at. Commit.